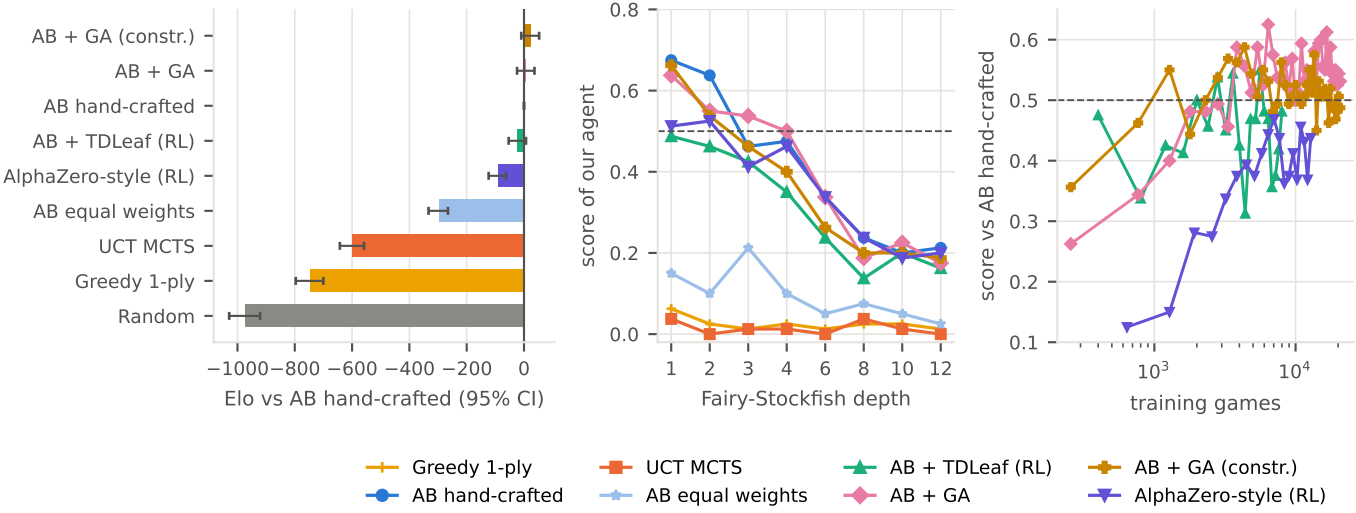
(a) E1: equal-budget round robin
AB + GA (constr.)
AB + GA
AB hand-crafted
AB + TDLeaf (RL)
AlphaZero-style (RL)
AB equal weights
UCT MCTS
Greedy 1-ply
Random
−1000 −800 −600 −400 −200 0
Elo vs AB hand-crafted (95% CI)
(b) E2: external opponent
score of our agent
0.0 0.2 0.4 0.6 0.8
1 2 3 4 6 8 10 12
Fairy-Stockfish depth
(c) E3: learning curves
score vs AB hand-crafted
0.1 0.2 0.3 0.4 0.5 0.6
$10^3$ $10^4$
training games
Greedy 1-ply
AB hand-crafted
UCT MCTS
AB equal weights
AB + TDLeaf (RL)
AB + GA
AB + GA (constr.)
AlphaZero-style (RL)